

ESP32-S3-DevKitC-1 (N16R8) — FINAL PIN MAP

Pins 1–22 = LEFT header (pin 1 = 3V3, top) · Pins 23–44 = RIGHT header (pin 23 = first GND, top)

Pin #	GPIO / PIN	Connects To	PU / PD	Value	Comments
1	3V3	—	—	—	NC. DevKit LDO output — never back-feed
2	3V3	—	—	—	NC
3	RST (EN)	Test point	—	—	Optional reset button to GND
4	GPIO4	On-board divider tap	—	—	BATT_SENSE. ADC1_3. 100 k/20 k + 100 nF + BAT54S
5	GPIO5	CN2 pin 6	—	—	M2_ENC_B
6	GPIO6	J-PROX-L pin 2, via 10 k series	PU	4.7 kΩ	PROX_LEFT. ⚠ 4k7 SENSOR side
7	GPIO7	J-PROX-R pin 2, via 10 k series	PU	4.7 kΩ	PROX_RIGHT. ⚠ 4k7 SENSOR side
8	GPIO15	BNO086 pin 9 RST	PU	10 kΩ	IMU_RST, active low. R28
9	GPIO16	TCA9548A SDA · J-I2C pin 3	PU	2.2 kΩ	I2C_SDA
10	GPIO17	TCA9548A SCL · J-I2C pin 4	PU	2.2 kΩ	I2C_SCL
11	GPIO18	TCA9548A RESET	PU	10 kΩ	MUX_RST, active low
12	GPIO8	Q1 gate (2N7002), via 100 Ω	PD	10 kΩ	IR_EMIT. PD at the gate
13	GPIO3	—	—	—	NC. Strapping (JTAG sel)
14	GPIO46	—	—	—	NC. Strapping (LOG)
15	GPIO9	BNO086 pin 10 WAK	PU	10 kΩ	IMU_WAKE. R29 — shared node with PS0
16	GPIO10	BNO086 pin 11 CS	PU	10 kΩ	IMU_CS. FSPICS0. R27
17	GPIO11	BNO086 pin 12 SI	—	—	IMU_MOSI. FSPID
18	GPIO12	BNO086 pin 14 SCK	—	—	IMU_SCLK. FSPICLK. Symbol says SDK
19	GPIO13	BNO086 pin 13 SO	—	—	IMU_MISO. FSPIQ
20	GPIO14	BNO086 pin 8 INT	—	—	IMU_INT, active low
21	5V	Board 5 V rail via D5	—	—	BUCK#2 5.09 V → SS34 → 4.74 V
22	GND	GND plane	—	—	
23	GND	GND plane	—	—	
24	GND	GND plane	—	—	
25	GPIO19	—	—	—	NC. USB D–
26	GPIO20	—	—	—	NC. USB D+
27	GPIO21	DRV8870 #1 IN2	PD	10 kΩ	M1_PWM. ⚠ Mandatory

Pin #	GPIO / PIN	Connects To	PU / PD	Value	Comments
28	GPIO47	CN2 pin 5	—	—	M2_ENC_A
29	GPIO48	DRV8870 #1 IN1	PD	10 kΩ	M1_DIR. ⚠ Mandatory
30	GPIO45	—	—	—	NC. Strapping (VDD_SPI)
31	GPIO0	—	—	—	NC. BOOT button / strapping
32	GPIO35	—	—	—	NC. Octal PSRAM (R8)
33	GPIO36	—	—	—	NC. Octal PSRAM (R8)
34	GPIO37	—	—	—	NC. Octal PSRAM (R8)
35	GPIO38	Onboard RGB LED	—	—	STATUS_LED. No board wire. <code>neopixelWrite()</code>
36	GPIO39	CN1 pin 5	—	—	M1_ENC_A
37	GPIO40	CN1 pin 6	—	—	M1_ENC_B
38	GPIO41	DRV8870 #2 IN2	PD	10 kΩ	M2_PWM. ⚠ Mandatory
39	GPIO42	DRV8870 #2 IN1	PD	10 kΩ	M2_DIR. ⚠ Mandatory
40	GPIO2	U-AMP2 output	—	—	M2_ISENSE. ADC1_1. 1 kΩ + 1 μF
41	GPIO1	U-AMP1 output	—	—	M1_ISENSE. ADC1_0. 1 kΩ + 1 μF
42	GPIO44	J-CONSOLE pin 3	—	—	U0RXD (silkscreen RX)
43	GPIO43	J-CONSOLE pin 2	—	—	U0TXD (silkscreen TX)
44	GND	GND plane	—	—	

44 pins · 26 GPIO carry a board net · GPIO38 driven on the DevKit itself · zero spares.

9 GPIO deliberately NC: 0, 3, 19, 20, 35, 36, 37, 45, 46 — plus both 3V3 header pins.

Two things worth noting from re-deriving this:

The GND positions are your mirror check. Pins 22, 23, 24 and 44 are all GND — two at the top of J3, one at the bottom. If your footprint's pad numbering doesn't put GNDs in exactly those four positions, the header is mirrored.

The count was off by one. The doc said "25 assigned"; it's 26 GPIO carrying a board net. GPIO38 was being double-counted against the spare tally. Corrected.

The pull-up/pull-down rationale from my last message stands unchanged — nothing there moved, only the row ordering and numbering. §3.1 now carries this exact table.